Phospholipids in rice: significance in grain quality and health benefits: a review.

Phospholipids (PLs) are a major class of lipid in rice grain. Although PLs are only a minor nutrient compared to starch and protein, they may have both nutritional and functional significance. We have systemically reviewed the literature on the class, distribution and variation of PLs in rice, their relation to rice end-use quality and human health, as well as available methods for analytical profiling. Phosphatidylcholine (PC), phosphatidylethanolamine (PE), phosphatidylinositol (PI) and their lyso forms are the major PLs in rice. The deterioration of PC in rice bran during storage was considered as a trigger for the degradation of rice lipids with associated rancid flavour in paddy and brown rice. The lyso forms in rice endosperm represent the major starch lipid, and may form inclusion complexes with amylose, affecting the physicochemical properties and digestibility of starch, and hence its cooking and eating quality. Dietary PLs have a positive impact on several human diseases and reduce the side-effects of some drugs. As rice has long been consumed as a staple food in many Asian countries, rice PLs may have significant health benefits for those populations. Rice PLs may be influenced both by genetic (G) and environmental (E) factors, and resolving G×E interactions may allow future exploitation of PL composition and content, thus boosting rice eating quality and health benefits for consumers. We have identified and summarised the different methods used for rice PL analysis, and discussed the consequences of variation in reported PL values due to inconsistencies between methods. This review enhances the understanding of the nature and importance of PLs in rice and outlines potential approaches for manipulating PLs to improve the quality of rice grain and other cereals.